

First Tool: Wireshark

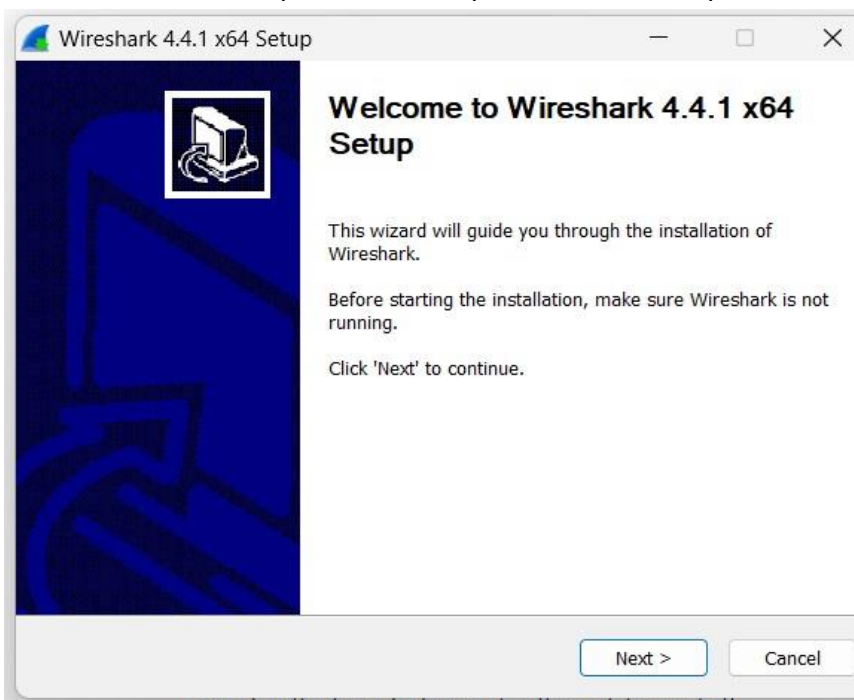
Visit the Wireshark Website

Go to the official Wireshark download page:<https://www.wireshark.org/download.html> Choose the **Windows Installer (64-bit)** option to begin the download.



Once the download is complete, locate the downloaded .exe file and double-click it to run the installer.

The Wireshark Setup Wizard will open. Click **Next** to proceed through the installation process.

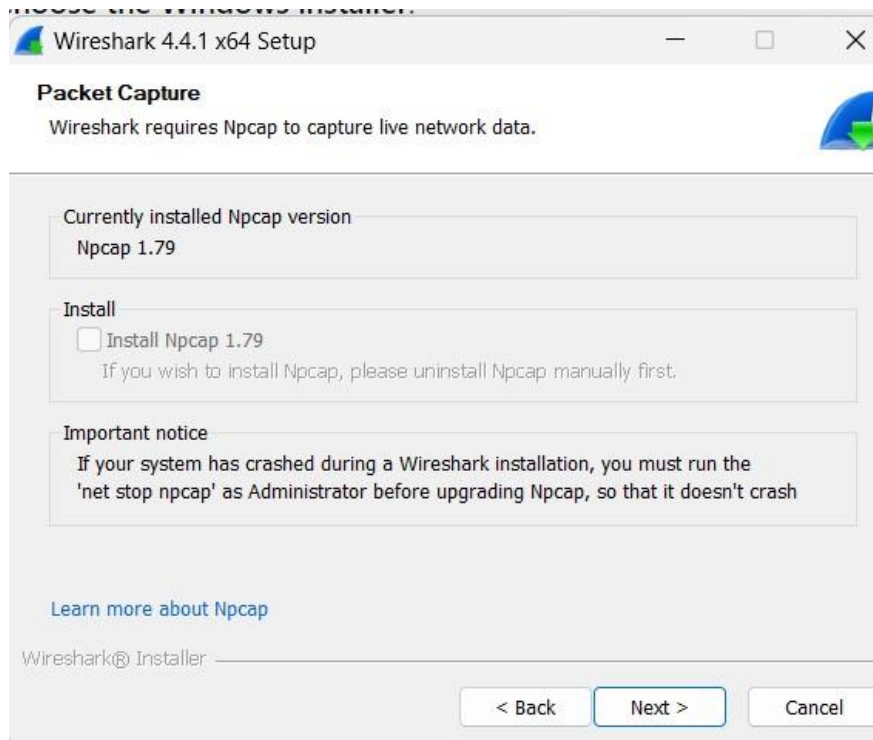


Continue clicking **Next** until you reach the option to choose the installation location. Select your preferred location and click **Next**.

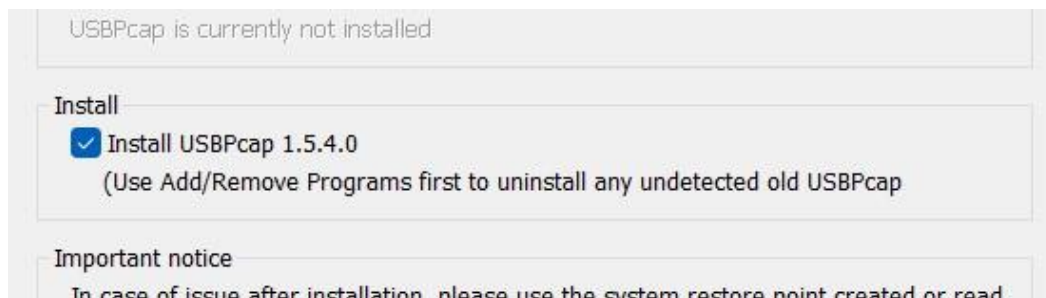
During the installation, you will be prompted to install **WinPcap**, which is used for packet capturing. If you do not already have it installed, allow it to install.

I already have Npcap 1.79, so there's no need to download it. But if you don't have it, make sure to check the option to download it.

If you have **Npcap** version 1.79 or later, you can skip this step. Otherwise, ensure the option to download and install Npcap is checked.



Check the box for **USBPcap 1.5.4** to include it in the installation.



After following the above steps, click **Finish** once the installation is complete. Your system may need to restart.

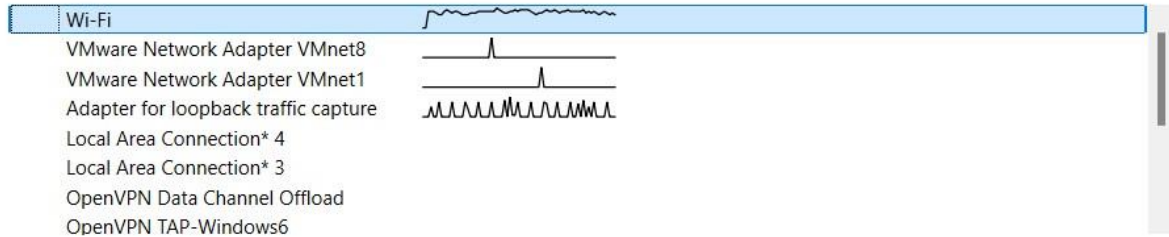
Capturing Packets

Launch Wireshark from your Start Menu or desktop shortcut.

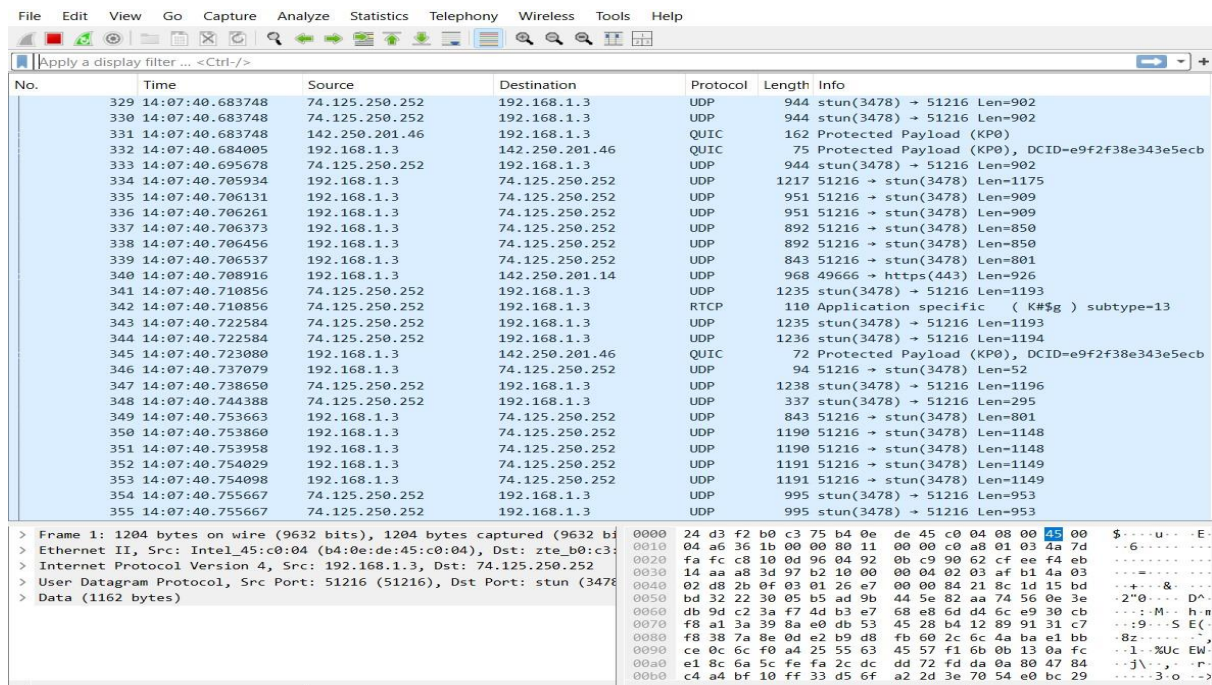
Once opened, Wireshark will begin capturing packets automatically. You can select packets transmitted over Wi-Fi.

Capture

...using this filter: All interfaces shown ▾

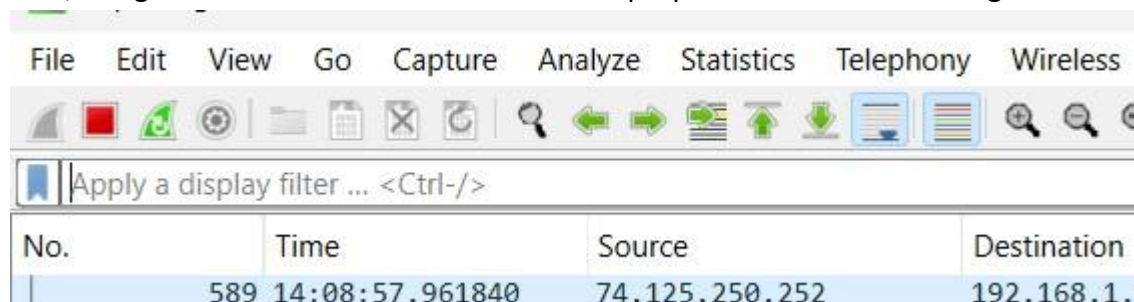


Learn



To save the captured packets in a .pcap file, you must first stop the capture by clicking the red stop button.

Then, navigate to **File** → **Save As** and select the .pcap file extension for saving.



Then file ⑦ save as ⑦ choose pcap extension

File name: network

Save as: Wireshark/tcpdump/... - pcap

Compression options

☒ Uncompressed

Second Tool: Nessus Tool Installation

Steps

Go to the official Nessus download page: <https://www.tenable.com/products/nessus/nessus-essentials>

Enter your information to register for Nessus. After completing the registration, check your email for an activation code. Save this code for later use.



Thank you for registering for Nessus® Essentials. An email containing your activation code has been sent to you at the email address you provided.

Download Nessus

To download Nessus, visit the Nessus Download page.

[Download](#)

Select the appropriate version of Nessus for your system and begin the download.

The image shows a screenshot of the "Tenable Nessus" website's download page. The header is "Tenable Nessus" in a bold, dark font. Below the header, there's a section titled "1 Download and Install Nessus". Under this, it says "Choose Download". There are two dropdown menus: "Version" with "Nessus - 10.8.3" selected, and "Platform" with "Windows - x86_64" selected. Below these is a blue button with a download icon and the text "Download". To the right of the button is a link for "Checksum". At the bottom of this section is a link that says "Download by curl >".

Once the download is complete, double-click the installer to launch the Tenable Nessus.



Click **Next**.

Accept the license agreement and click **Next** again.

Choose your preferred installation location and click **Next**.

Click **Install** to begin the installation process.

Once installed, Nessus will start running as a service.

Once the installation is complete, click **Finish**. Nessus will start running as a service automatically.

Accessing the Nessus Web Interface

Navigate to <https://localhost:8834/> to access the Nessus web interface. If the interface does not open automatically, you can enter this URL manually.

When prompted with a "Connection is not private" warning, click Advanced, then select Proceed to localhost.



Connect via SSL

NOTICE: If you get a security alert from your browser, you can accept the risk and continue or obtain a valid certificate before proceeding. Please refer to the documentation for more information.



Your connection is not private

Attackers might be trying to steal your information from **localhost** (for example, passwords, messages, or credit cards). [Learn more about this warning](#)

NET::ERR_CERT_AUTHORITY_INVALID

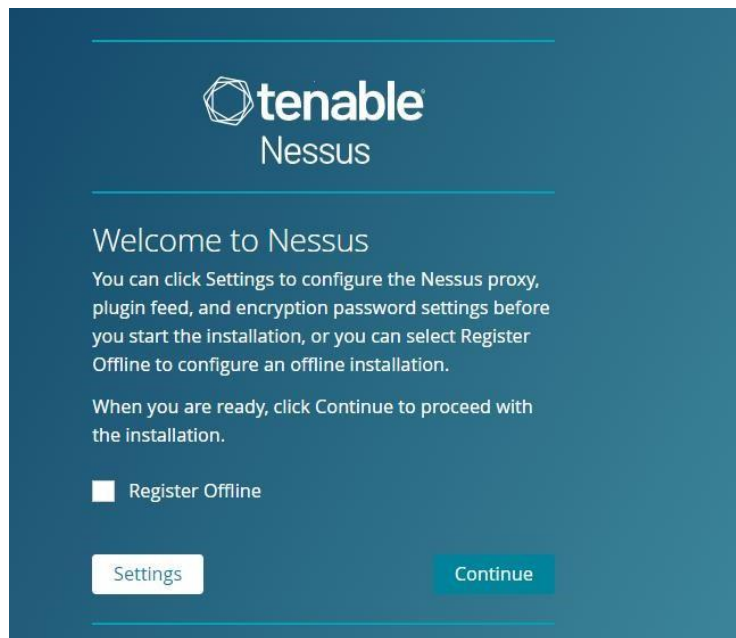
Hide advanced

Back to safety

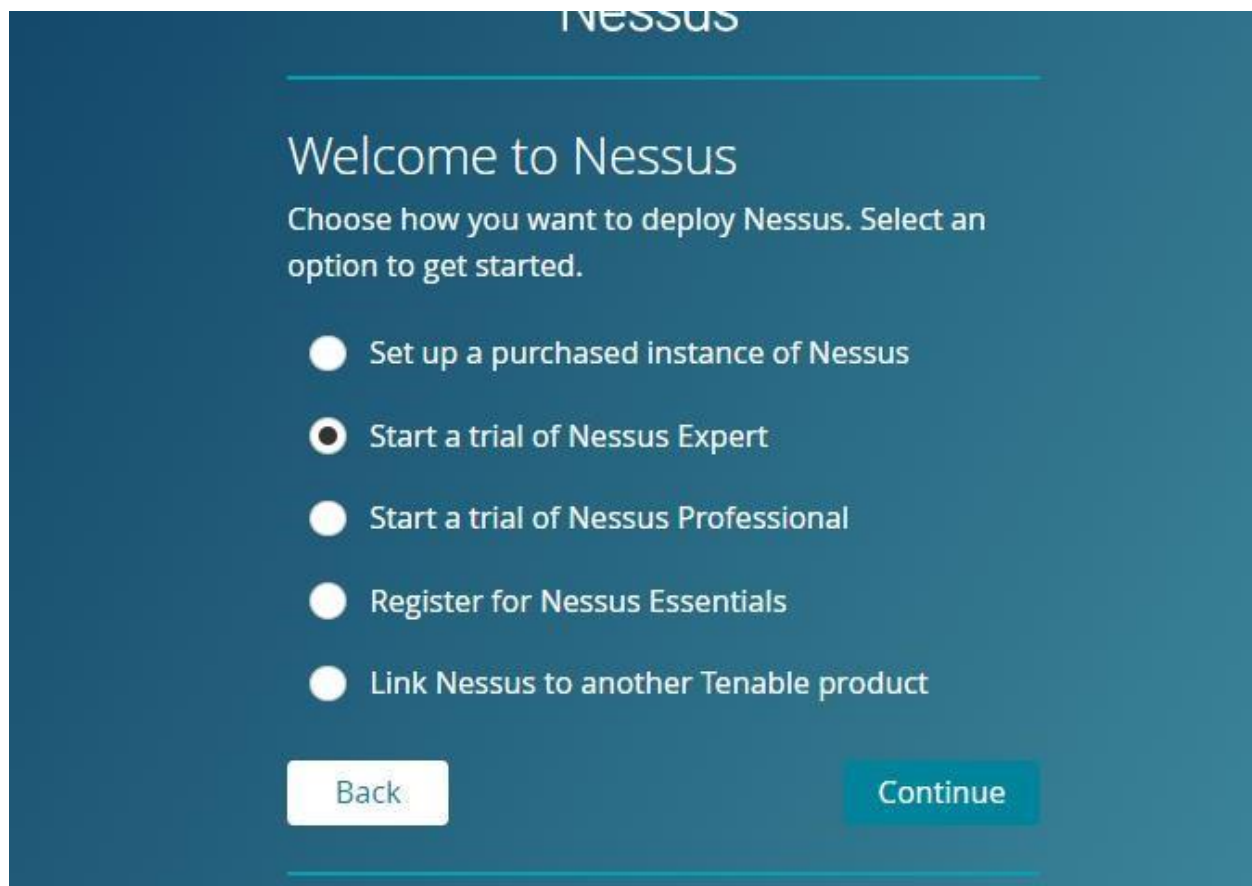
This server could not prove that it is **localhost**; its security certificate is not trusted by your computer's operating system. This may be caused by a misconfiguration or an attacker intercepting your connection.

[Proceed to localhost \(unsafe\)](#)

Press continues



Then start trial of nessus expert, then continue



You can skip the email registration step, as you have already completed it.

Enter the activation code that was sent to your email and click **Continue**.

Click **Continue** again when prompted.



Get Started

To start your Nessus Expert trial, please provide your business email address.

Email

Already have activation code? Skip this step to enter it manually.

Back

Skip

Continue

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Register Nessus

To find your activation code, please sign into your [Tenable Community](#) account or set a password to finish setting up your account.

Activation Code

Having an issue activating a product? Please open a technical support case on the [Tenable Community Portal](#).

Back

Continue

Press continue and continue again

Create a username and password for accessing Nessus



Create a user account

Create a Nessus administrator user account. Use this username and password to log in to Nessus.

Username *

Password *

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Then let it to download plugins



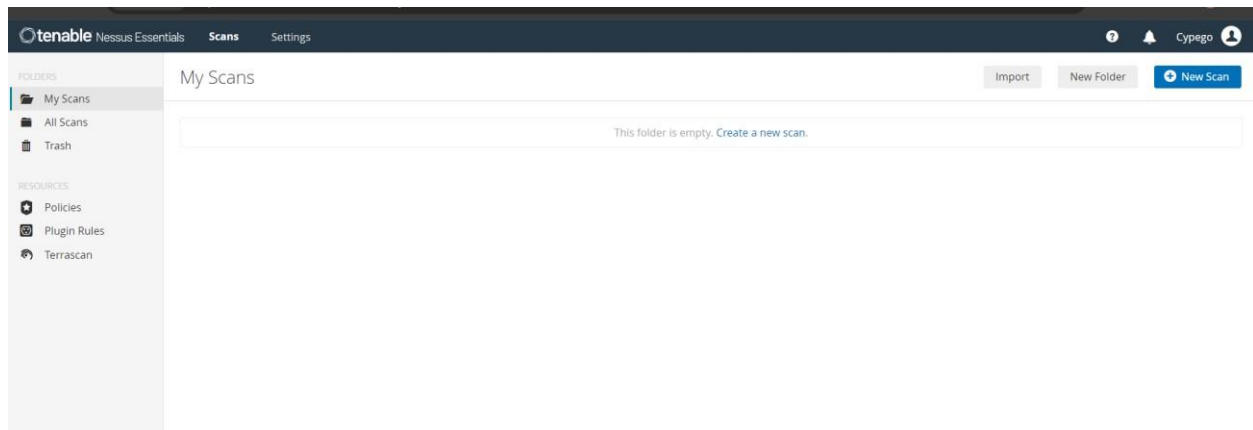
Initializing

Please wait while Nessus is initializing.

Downloading plugins...

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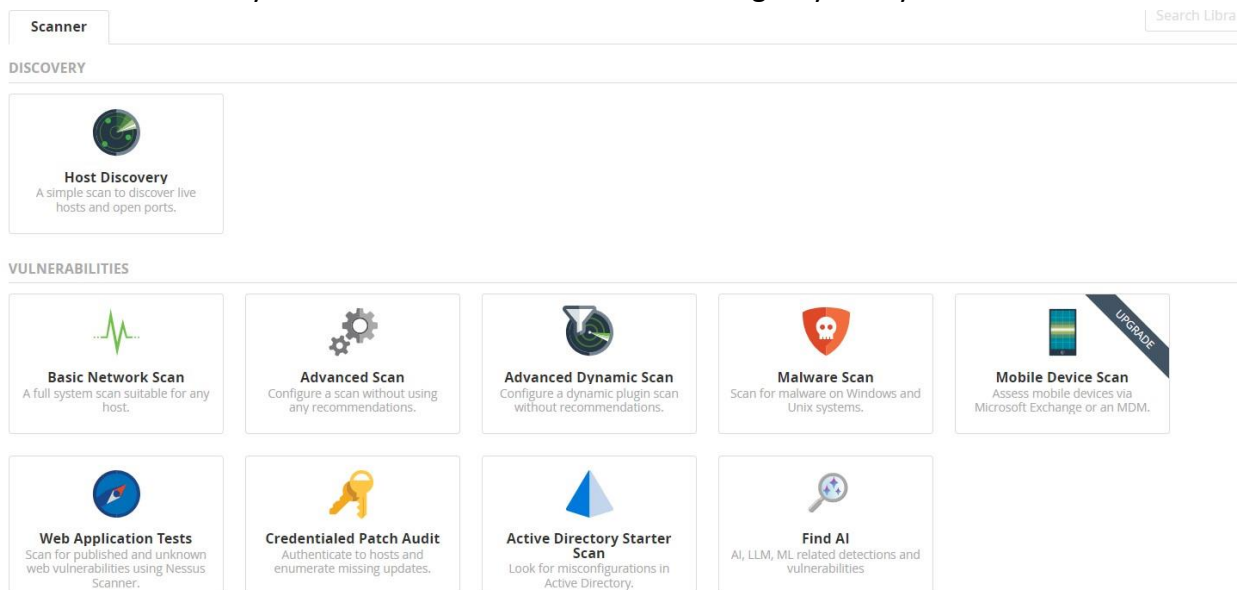
After the plugins are installed, Nessus will be ready to use.



Running Scans

From the Nessus dashboard, you can create a new scan. Choose any scan type that suits your needs.

Provide the necessary details and information about the target system you wish to scan.



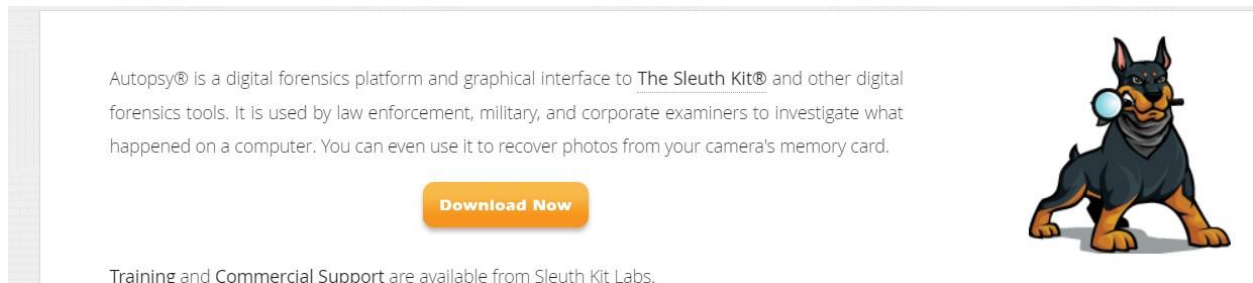
You can run scans anytime by navigating to <https://localhost:8834/>

Third Tool: Autopsy Tool

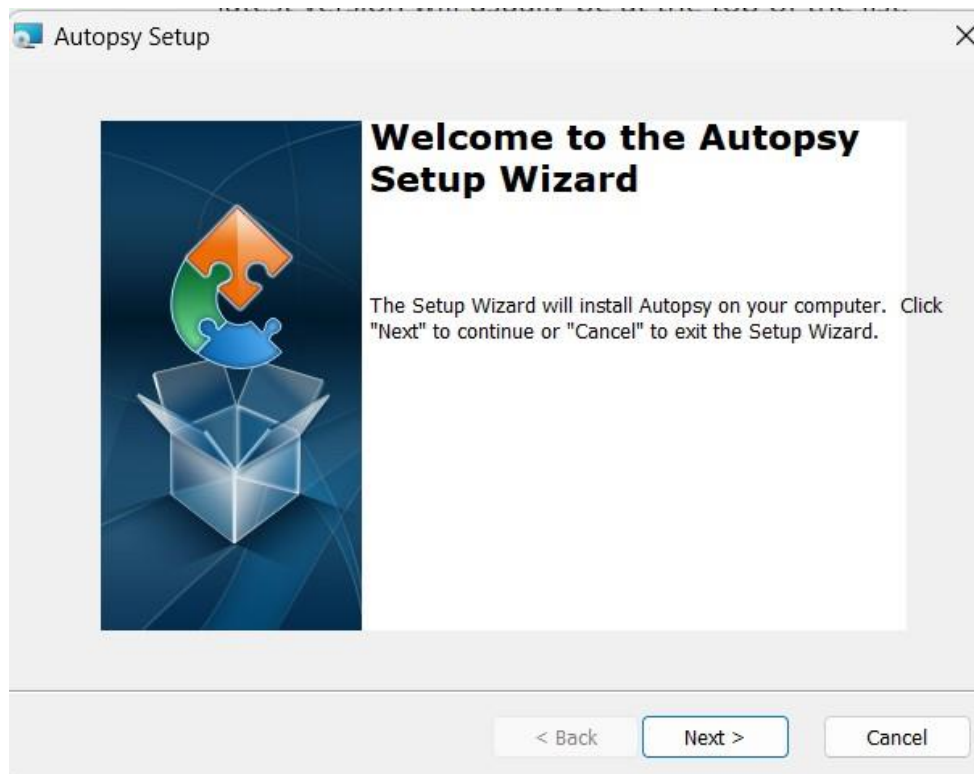
Visit the Autopsy Website

Go to the official Autopsy download page: <https://www.sleuthkit.org/autopsy/> Click on

Download Now and select the appropriate version for Windows.



After the download is complete, locate the installer and double-click it to set up Autopsy.

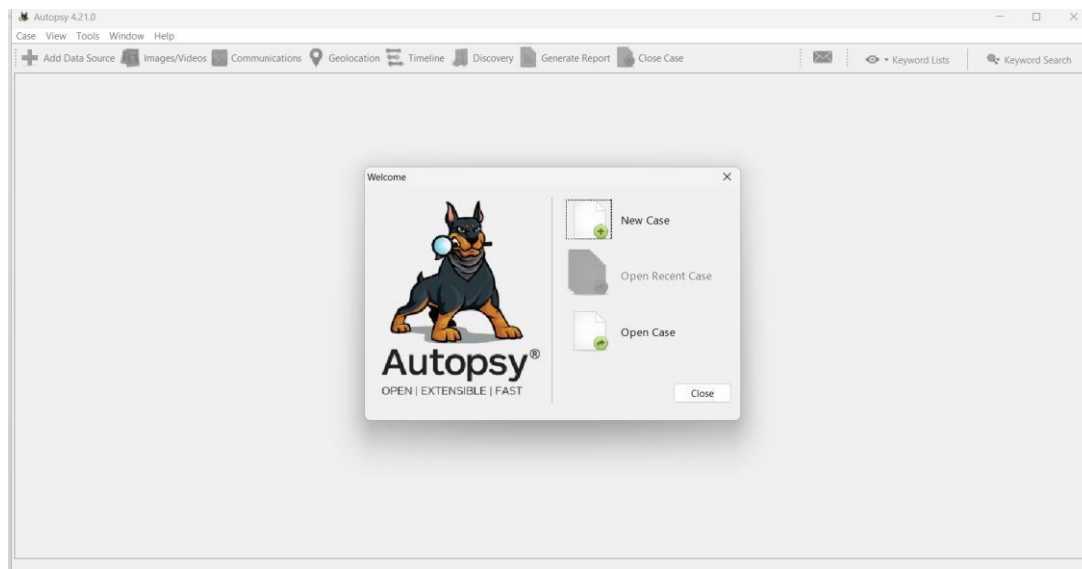


Click **Next** to proceed through the setup wizard.

Choose your preferred installation location and click **Next**.

Click **Install** to begin the installation process.

Once the installation is finished, you will have the option to create a new case or open an existing one.



Prerequisites • Java Requirement

Ensure that you have Java installed on your system, as Autopsy requires it to run. If you do not have Java installed, you can download it from the following link:

<https://www.oracle.com/java/technologies/downloads/#java11?er=221886>

Fourth Tool: Suricata

Installation Steps

Go to the official Suricata download page: <https://suricata.io/download/>

Look for the Windows installation package and ensure you select the appropriate version for your system architecture (32-bit or 64-bit).



Suricata requires Npcap for packet capturing. If you do not have it installed, download it from the following link:

<https://npcap.com/#download>

Run the Installer: Double-click the downloaded installer file to launch the installation wizard.

Follow the Installation Instructions:

- Accept the license agreement.
- Choose the installation directory (the default location is usually fine).
- Select any additional components you want to install, if prompted.
- Click "Install" to begin the installation.
- Then finish the installation



Configuration Steps

Navigate to C:\Program Files\Suricata\.

open the suricata.yaml file in a text editor or Visual Studio.

Run ipconfig in the command prompt to find your IP address and subnet mask.

Locate the HOME_NET variable in the suricata.yaml file and set it to match your internal network.

Example: If your network is 192.168.1.0/24 (where your device has the IP 192.168.1.3), then set HOME_NET to 192.168.1.0/24.

```
vars:
  # more specific is better for alert accuracy and performance
  address-groups:
    HOME_NET: "[192.168.1.0/24, 192.168.116.0/24, 192.168.136.0/24]"
    #HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[10.0.0.0/8]"
    #HOME_NET: "[172.16.0.0/12]"
```

Ensure that any rules you want to use are not hashed (commented out) in the configuration file.

Also, ensure that the EXTERNAL_NET variable is not hashed.


```
default-rule-path: C:\\Program Files\\Suricata\\rules\\

rule-files:
- botcc.rules
- botcc.portgrouped.rules
- ciarmy.rules
- compromised.rules
- drop.rules
- dshield.rules
- emerging-activex.rules
- emerging-adware_pup.rules
- emerging-attack_response.rules
- emerging-chat.rules
- emerging-coinminer.rules
- emerging-current_events.rules
- emerging-dns.rules
```

Also make sure externalnet is not hashed

```
EXTERNAL_NET: "!$HOME_NET"
#EXTERNAL_NET: "any"
```

Verify and Fix Rules

- Suricata uses rules to detect suspicious activity on the network. The rules are located in C:\Program Files\Suricata\rules.
- You can download additional rules from :
<https://rules.emergingthreats.net/open/suricata/>

 emerging-all.rules.zip.m3u	2024-10-16T20:11:20Z	33 B
 emerging.rules.tar.gz	2024-10-16T20:11:20Z	4.33 MB

Ensure that any new rules are properly declared in the suricata.yaml file.

Start Suricata

Open a command prompt and navigate to the Suricata installation directory.

To start Suricata with your IP address, enter the appropriate command in the command prompt. The logs will be stored in the log folder within the Suricata directory.

```
suricata.exe -c suricata.yaml -i 192.168.1.3
```

hint:

To test if your Suricata configuration is correct, run:

```
suricata.exe -T -c suricata.yaml
```

Fifth Tool: Volatility3

Navigate to the Volatility Repository

Visit the official Volatility 3 repository on GitHub

<https://github.com/volatilityfoundation/volatility3>

Click on the **Code** button, then select **Download ZIP**.

Once the ZIP file is downloaded, extract it to your preferred directory.

```
https://github.com/volatilityfoundation/volatil
```

Clone using the web URL.

 Open with GitHub Desktop

 Download ZIP

Open a command prompt and navigate to the directory where you extracted Volatility 3. Run the following command to install the required dependencies

```
pip install -r requirements.txt
```

```
PS C:\WINDOWS\system32> cd C:\volatility3-develop
PS C:\volatility3-develop> pip install -r .\requirements.txt
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pefile>=2023.2.7 in c:\users\emanm\appdata\local\packages\pythonsoftw...
3.12_qbz5n2kfra8p0\localcache\local-packages\python312\site-packages (from -r .\requirements-minima...
4.8.26)
Requirement already satisfied: vara-python>=3.8.0 in c:\users\emanm\appdata\local\packages\pythonsof...
```

To verify the installation and view available commands, run:

```
python vol.py -h
```

```
Volatility: error: unrecognized arguments: -h
PS C:\volatility3-develop> python vol.py -h
Volatility 3 Framework 2.11.0
usage: volatility [-h] [-c CONFIG] [--parallelism [{processes,threads,off}]] [-e EXTEND] [-p PLUGIN_DIRS]
                  [-s SYMBOL_DIRS] [-v] [-l LOG] [-o OUTPUT_DIR] [-q] [-r RENDERER] [-f FILE] [--write-config]
                  [--save-config SAVE_CONFIG] [--clear-cache] [--cache-path CACHE_PATH] [--offline | -u URL]
                  [--filters FILTERS] [--hide-columns [HIDE_COLUMNS ...]] [--single-location SINGLE_LOCATION]
                  [--stackers [STACKERS ...]] [--single-swap-locations [SINGLE_SWAP_LOCATIONS ...]]
                  {banners.Banners,configwriter.ConfigWriter,frameworkinfo.FrameworkInfo,isfinfo.IsfInfo,layerwriter.La
erWriter,linux.bash.Bash,linux.capabilities.Capabilities,linux.check_afinfo.Check_afinfo,linux.check_creds.Check_creds,
linux.check_idt.Check_idt,linux.check_modules.Check_modules,linux.check_syscall.Check_syscall,linux.ebpf.EBPF,linux.elfs
Elfs,linux.envvars.Envvars,linux.iomem.IOMem,linux.keyboard_notifiers.Keyboard_notifiers,linux.kmsg.Kmsg,linux.library_li
t.LibraryList,linux.lsmmod.Lsmmod,linux.lsof.Lsof,linux.malfind.Malfind,linux.mountinfo.MountInfo,linux.netfilter.Netfilt
r,linux.pagecache.Files,linux.pagecache.InodePages,linux.pidhashtable.PIDHashTable,linux.proc.Maps,linux.psaux.PsAux,li
```

Using Volatility 3 with Memory Dumps

To use Volatility with a memory dump, execute the following command

```
python vol.py -f <path_to_memory_dump> <plugin>
```

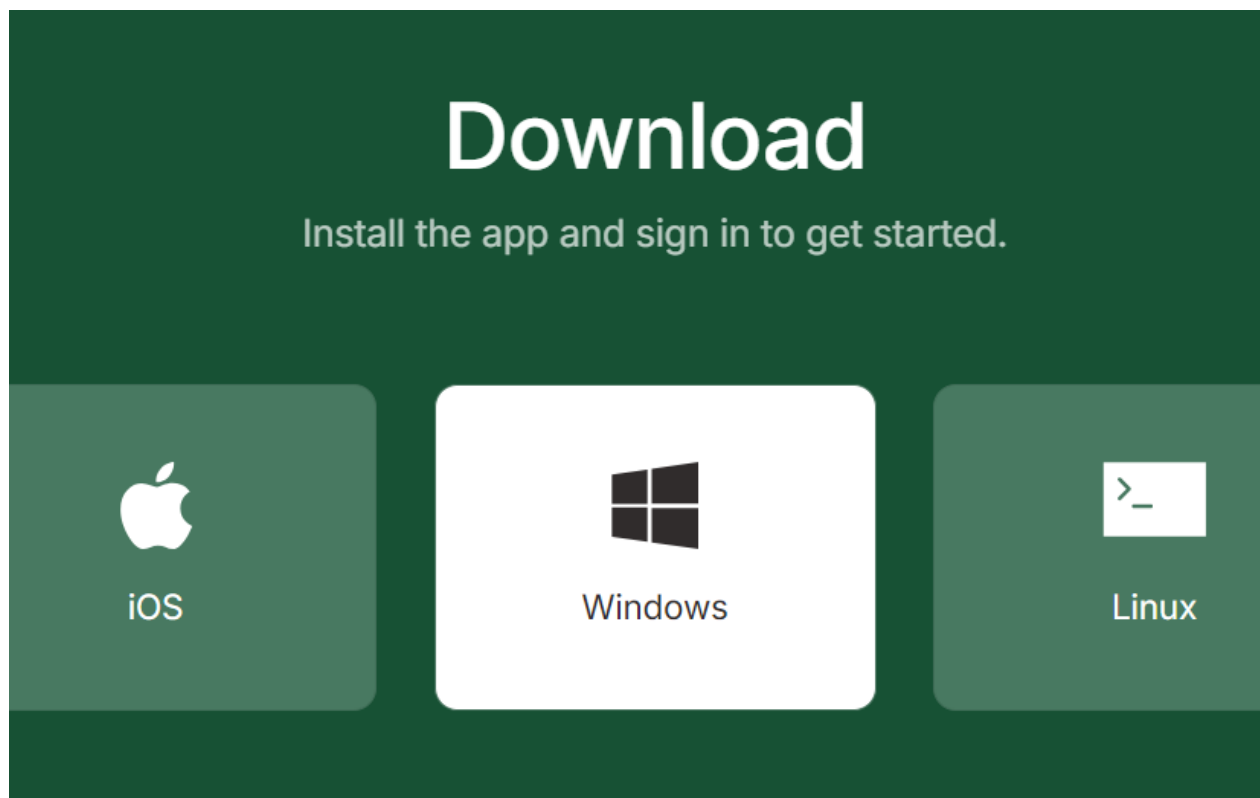
Replace <path_to_memory_dump> with the path to your memory dump file and <plugin> with the desired analysis plugin.

Sixth Tool: TailScale VPN

Download TailScale

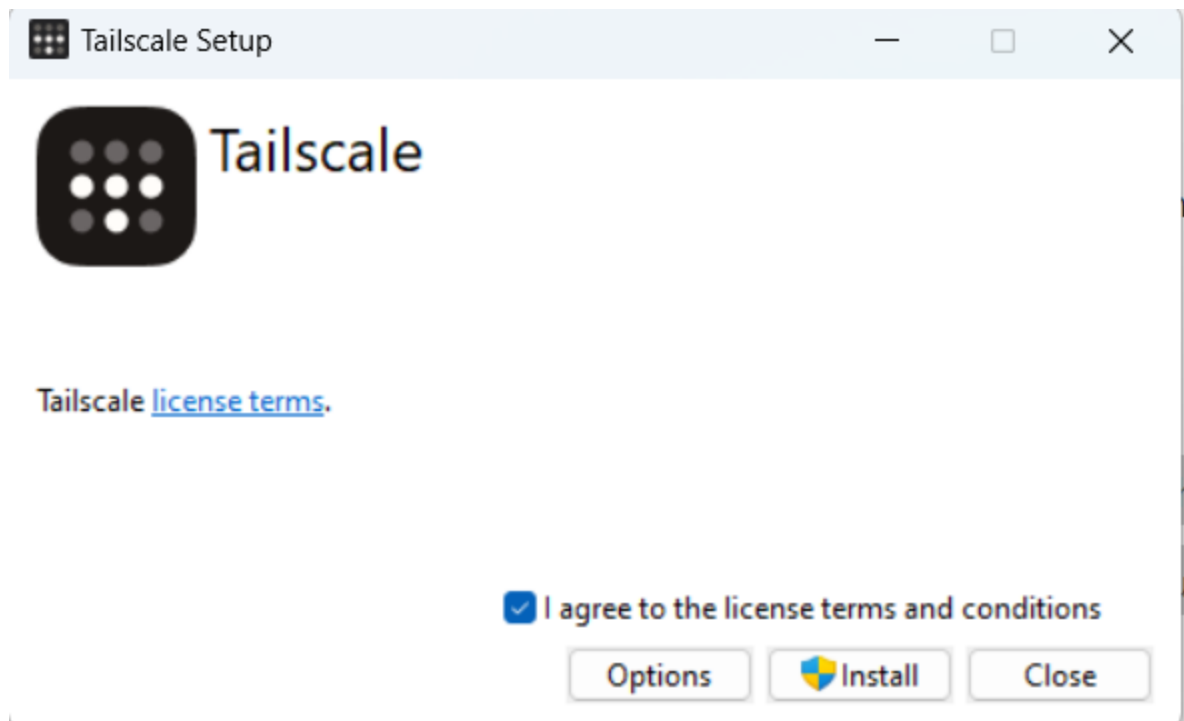
- Visit the link and click the "Download" button. Select the **Windows** option to download the installer.

<https://tailscale.com/download>



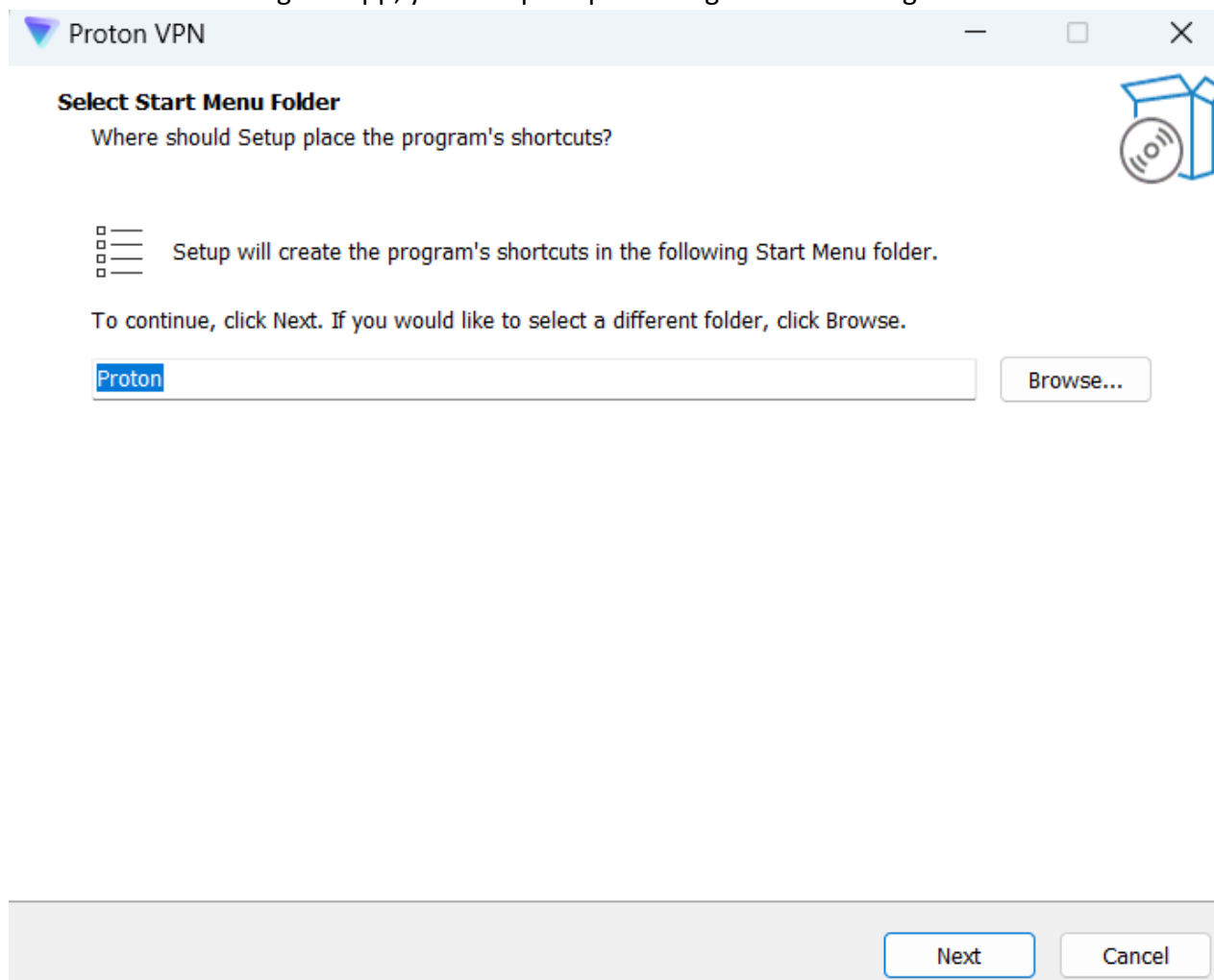
Install TailScale

- Run the installer and follow the instructions to complete the installation process.
- Once installed, open the TailScale app from the Start Menu.



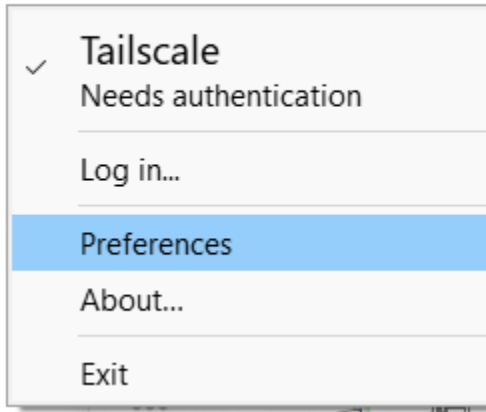
Sign In

- After launching the app, you'll be prompted to sign in. You can sign in with Gmail



Sign In

- After launching the app, you'll be prompted to sign in. Tailscale supports multiple sign-in methods
- Try to open it from start menu, if do not appear you will find it in the bottom right corner in system tray
- Select press login



Choose method



Log in to connect a device to your tailnet.

Enter your email...

Sign in

OR



Sign in with Google



Sign in with Microsoft



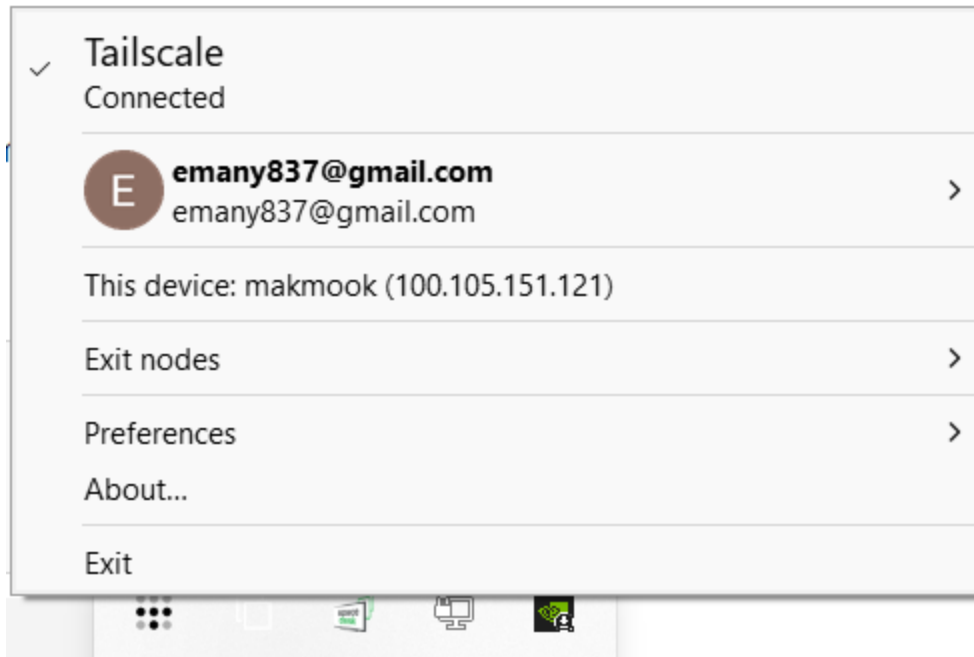
Sign in with GitHub



Sign in with Apple

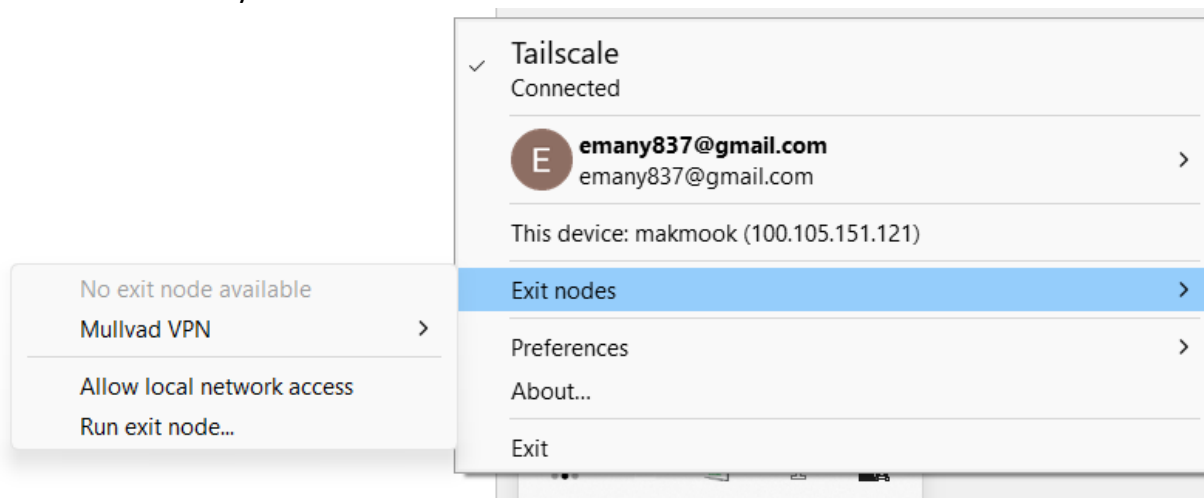


Sign in with a passkey



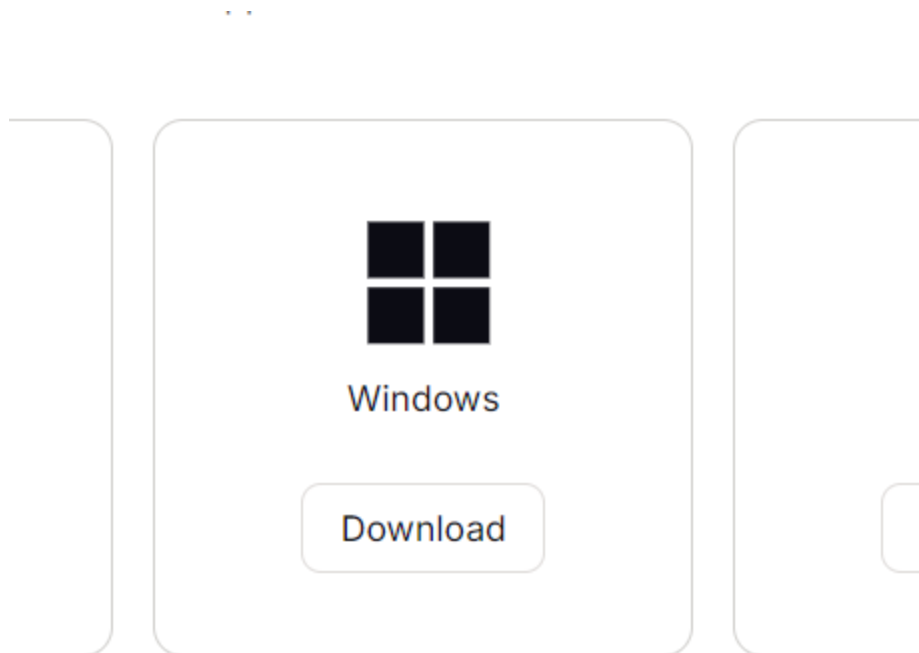
Set Up the Exit Node

- On the device you want to use as the exit node:

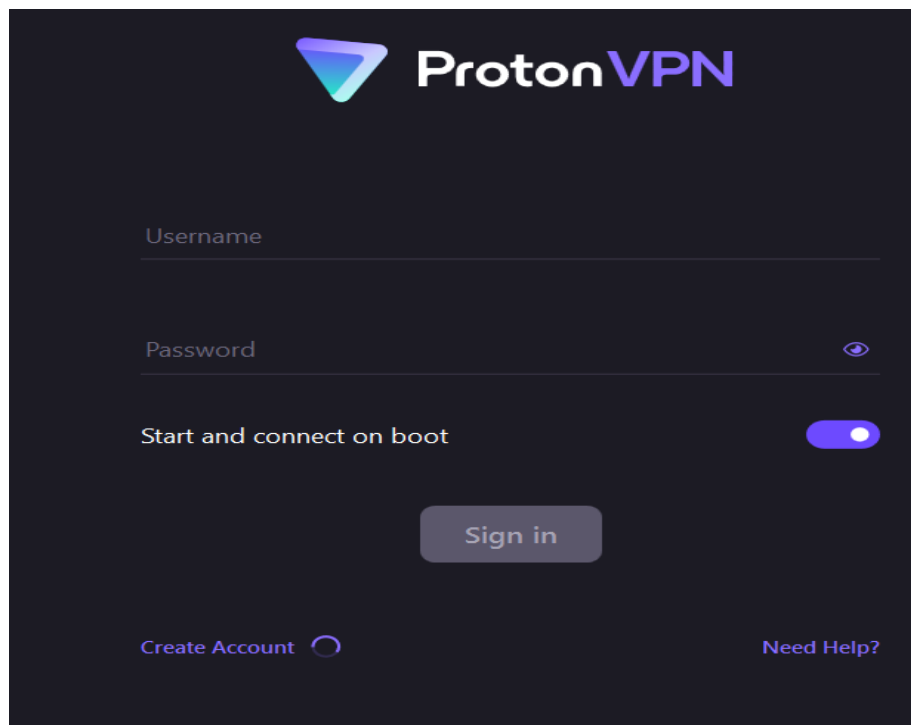


Another VPN(ProtonVPN)

Visit the ProtonVPN Website and download it



- Once the download is complete, run the installation.
- Follow the on-screen instructions to install ProtonVPN. You may need to accept the license agreement and choose the installation location.
- After installation, launch the ProtonVPN app. You will be prompted to sign in.
- If you don't have an account, you can create one directly from the app or on the ProtonVPN website.



Or [sign up for free](#)

- Once signed in, you'll see a list of available servers. Choose a server location and click Connect to start using ProtonVPN.

What Is My IP?

My Public IPv4: [37.19.200.25](#) 

My Public [IPv6](#): Not Detected

My IP Location: Dallas, TX US 

My ISP: DataCamp Limited 

Seventh Tool: Splunk on windows

- Go to Splunk download page https://www.splunk.com/en_us/download.html
- Select splunk enterprise

Splunk Enterprise

Download and install Splunk Enterprise trial on your own hardware or cloud instance so you can collect, analyze, visualize and act on all your data — no matter its source. Try indexing up to 500MB/day for 60 days, no credit card required.

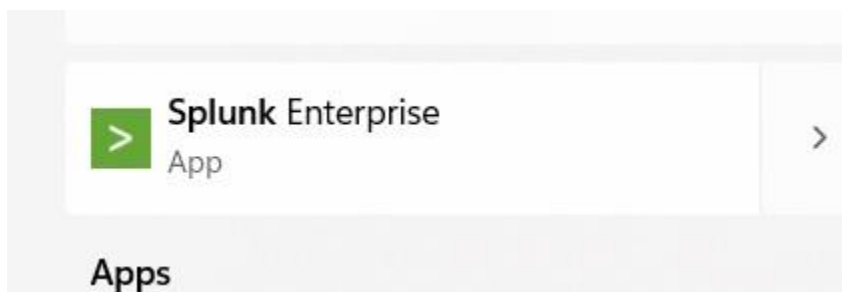
[Get My Free Trial](#)

[View Product](#)

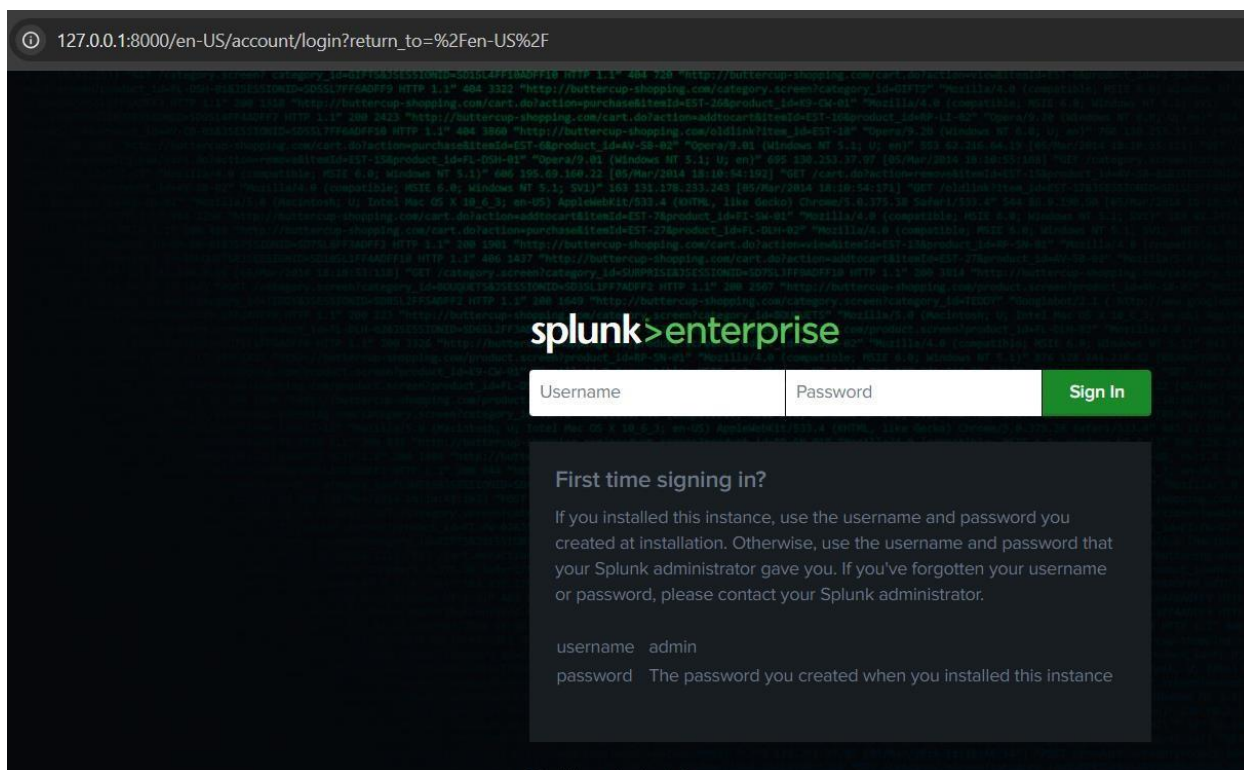
- Choose windows and download the msi
- Once downloaded, locate the .msi file and run the installer
- During installation, you'll be asked to set the installation directory, accept the license agreement, and choose to create shortcuts.

Set Splunk Admin Credentials:

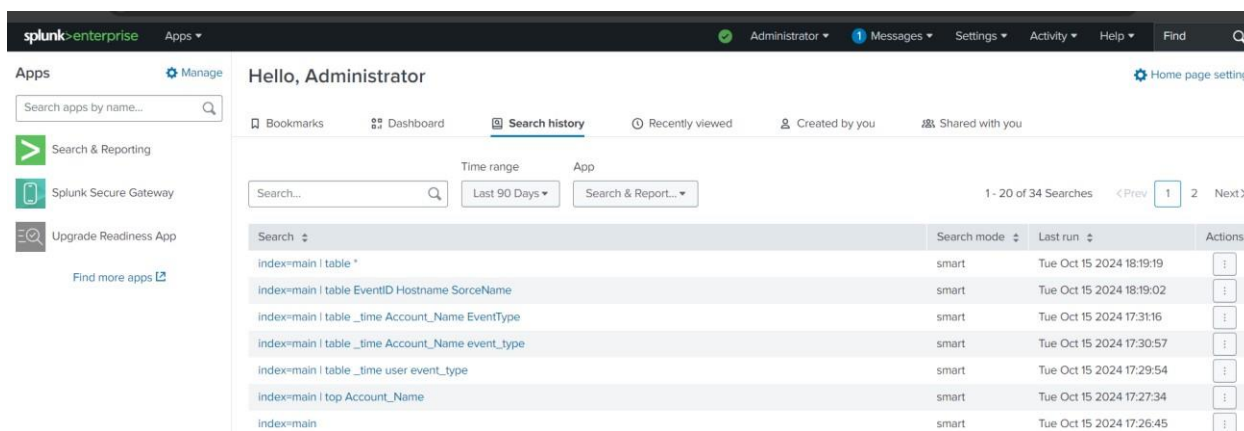
- You'll be prompted to set a username and password for the Splunk admin account during the installation. This will be used to log into Splunk's web interface. You can open it



Enter the username and password you have written



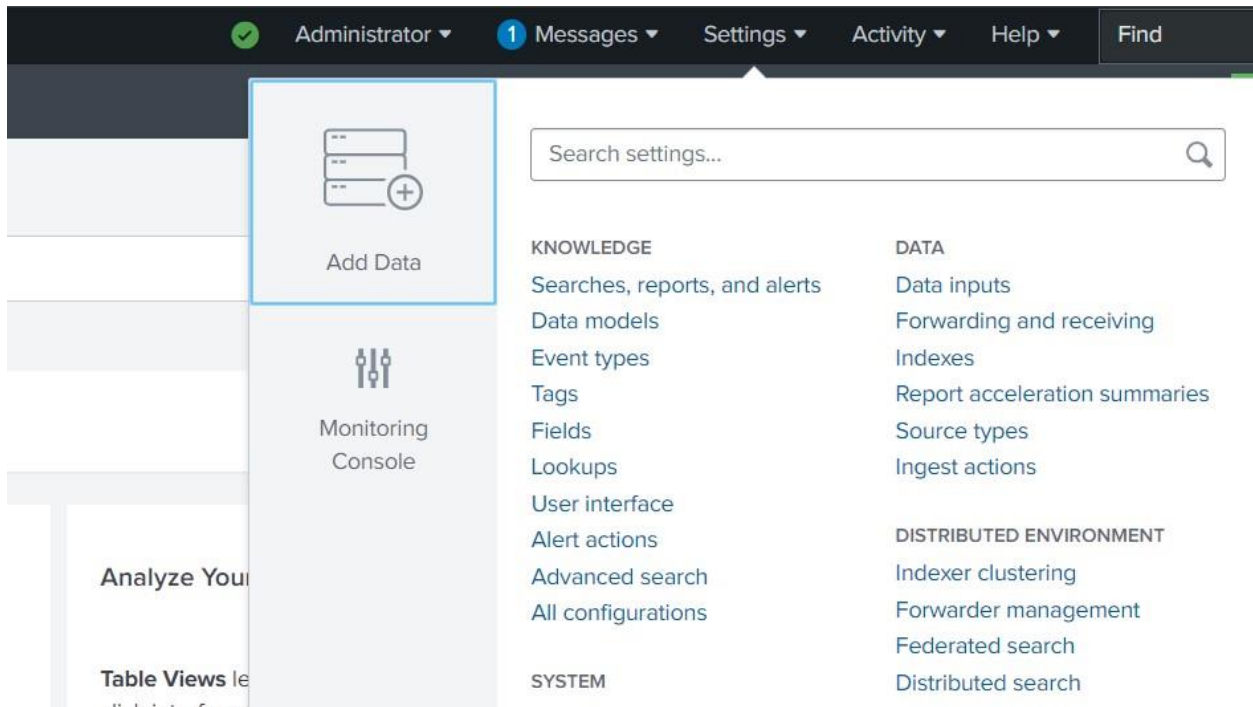
Then it will open



To add network logs

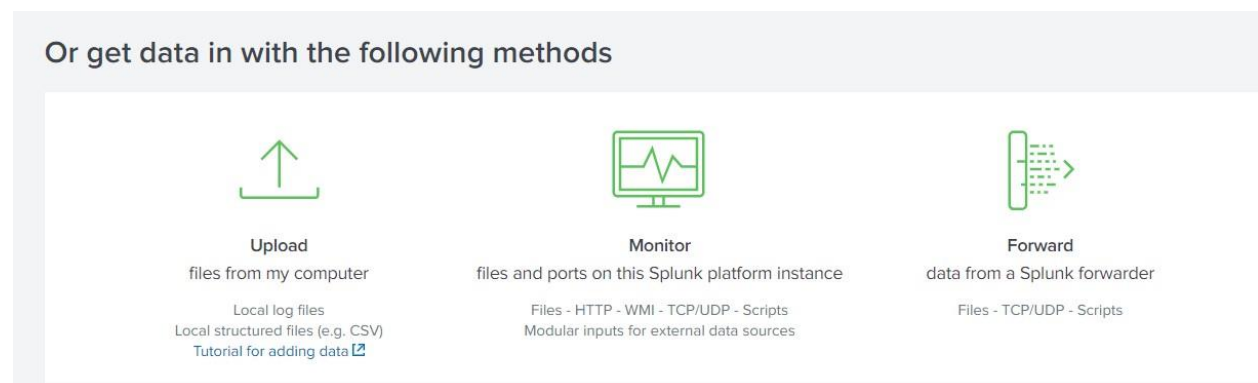
1. Add data

- In the Splunk home screen, click on "Settings" in the top-right corner.
- Under "Data", click "Add Data".



2. select data source

- You will see three options: Upload, Monitor, and Forward.
- Choose "Upload" if you want to upload a network log file manually



3. Upload the Network Log File:

- Click "Select File" and browse your computer for the network log file you want to upload.
- Once the file is selected, click "Next".

Selected File: **Netwok.pcap**

Select File

4. Source Type:

- Splunk will try to automatically detect the source type
- If Splunk does not correctly identify the source type, manually choose one that best fits your data

The screenshot shows the Splunk web interface for setting the source type. At the top, a progress bar indicates the current step is 'Set Source Type'. Below the progress bar, the title 'Set Source Type' is followed by a brief instruction: 'This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".' The 'Source' is listed as 'Netwok.pcap'. A dropdown menu for 'Source type' is set to 'Network'. There is a 'Save As' button and a table with columns for 'List', 'Format', and '20 Per Page'. A '< Prev' button is also visible.

- Click "Next" once the source type is selected.

5. Set Host, Index, and Sourcetype:

- **Host:** Identify the host from which the logs were generated. You can manually specify or let Splunk assign a host.
- **Index:** Select an index or create a new one to store your data. By default, you can use the "main" index.
- **Sourcetype:** Ensure the correct sourcetype is selected.
- After reviewing these settings, click "Review" and then "Submit".

The screenshot shows the 'Review' step in the Splunk data upload process. The progress bar at the top shows that 'Review' is the current step. The 'Review' section contains a summary of the configuration: Input Type: Uploaded File, File Name: Netwok.pcap, Source Type: Network, Host: MakMook, and Index: Default. At the bottom right, there are '< Back' and 'Submit >' buttons.

Data Uploaded:

- Your network log data will now be uploaded into Splunk and available for searching and analysis.

To add your logs

Search settings...

KNOWLEDGE

- Searches, reports, and alerts
- Data models
- Event types

DATA

- [Data inputs](#)
- Forwarding and receiving
- Indexes

Local inputs

Type	Inputs	Actions
Local event log collection Collect event logs from this machine.	-	Edit
Remote event log collections Collect event logs from remote hosts. Note: this uses WMI and requires a domain account.	1	+ Add new

Choose local event log collection and choose which logs you want to add then save

Name  localhost

Logs

Available log(s)

Add all »

Application ✓

Dell ✓

DirectShowFilterGraph

DirectShowPluginControl

Els_Hyphenation/Analytic

>

<

Selected log(s)

« Remove all

Application

Dell

ForwardedEvents

General Logging

HardwareEvents

Select the Windows Event Logs you want to index from the list.

i	Time	Event
>	10/15/24 2:51:31.000 PM	10/15/2024 02:51:31 PM LogName=Security EventCode=5379 EventType=0 ComputerName=MakMook SourceName=Microsoft Windows security auditing. Type=Information RecordNumber=5178886 Keywords=Audit Success TaskCategory=User Account Management OpCode=Info Message=Credential Manager credentials were read. Subject: Security ID: S-1-5-21-2554599862-2962239839-24097432-1001 Account Name: emanM Account Domain: MAKMOOK Logon ID: 0x13BF13A2 Read Operation: Enumerate Credentials This event occurs when a user performs a read operation on stored credentials in Credential Manager. Collapse host = MAKMOOK source = WinEventLog:Security sourcetype = WinEventLog:Security